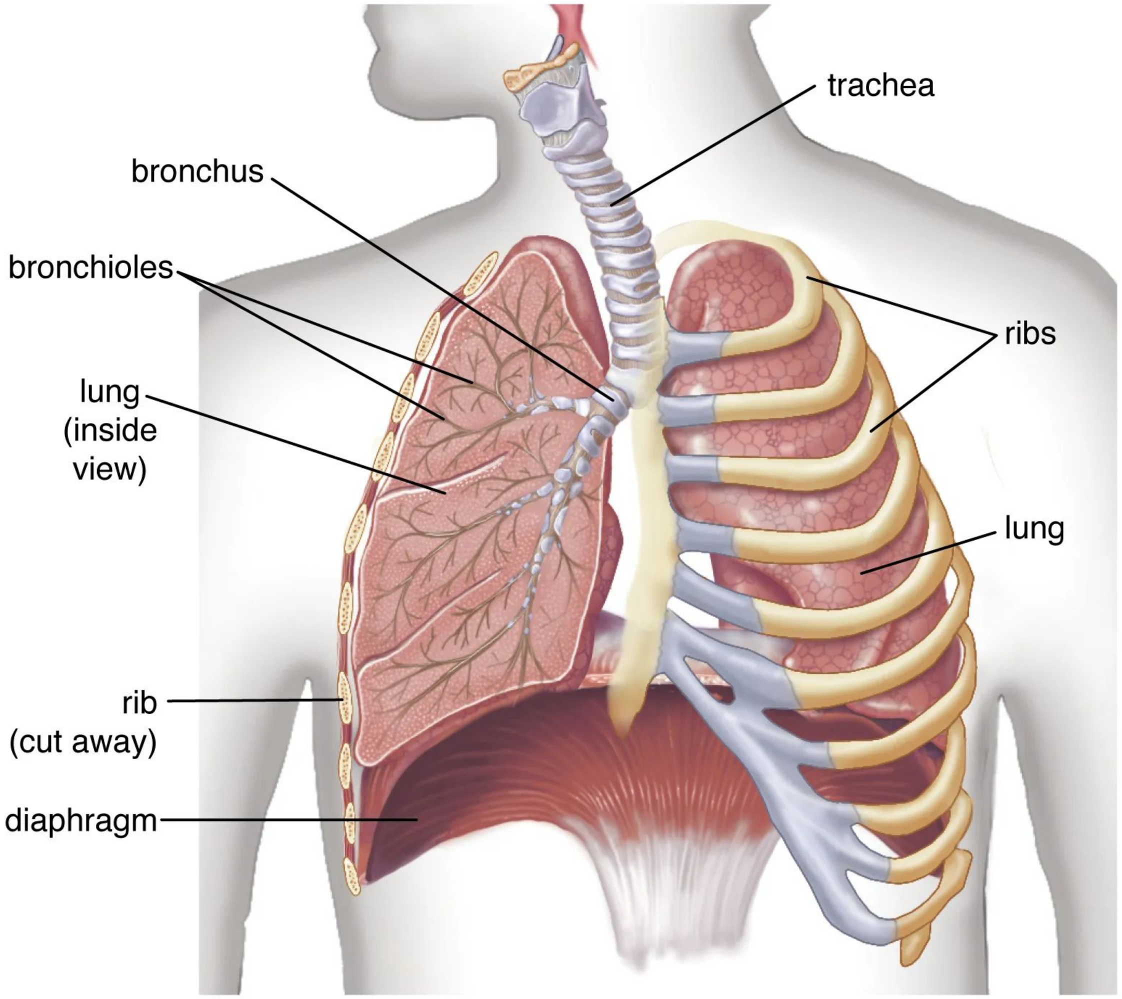
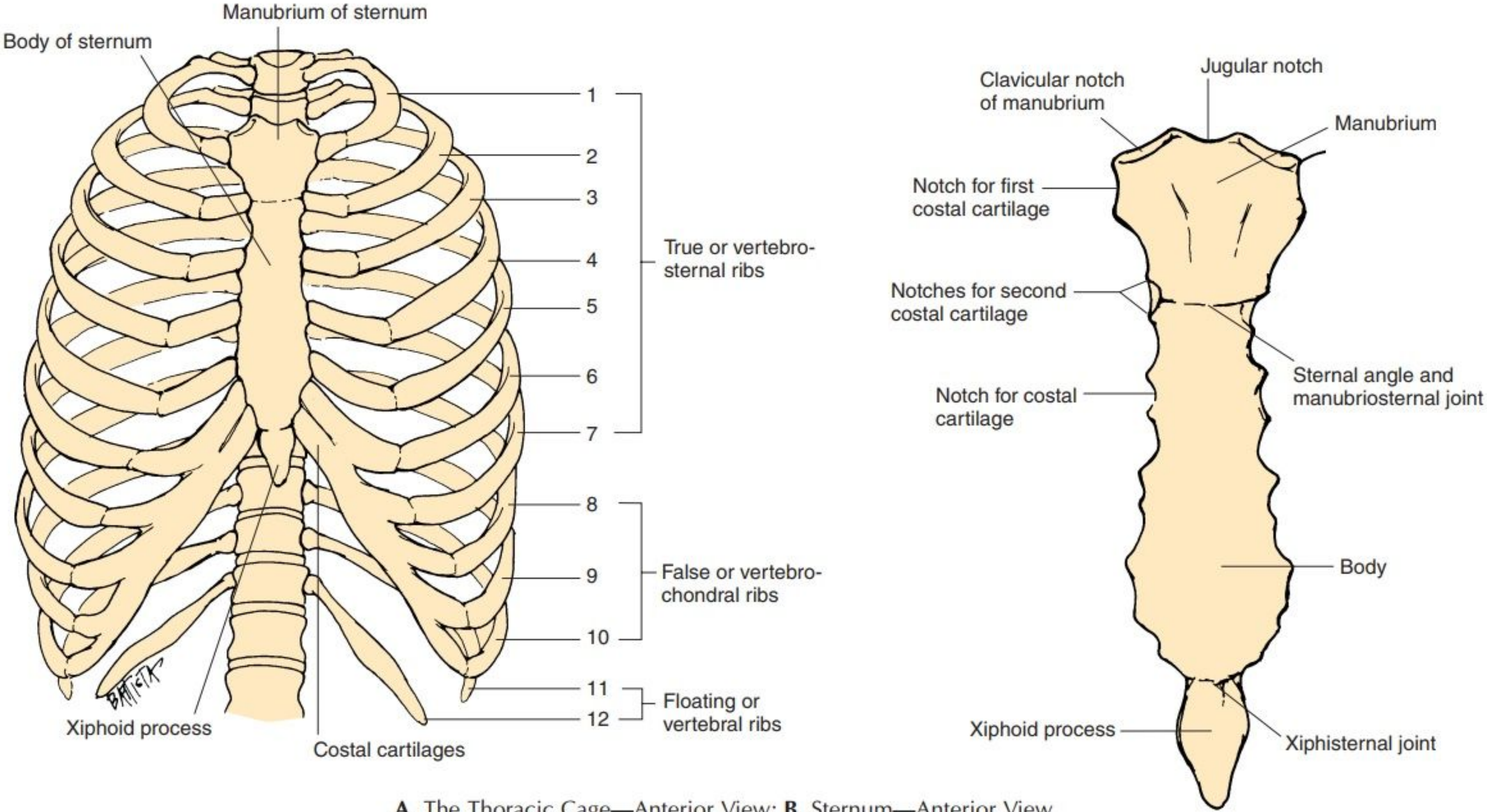
Chest X-ray Diseases

Chest Anatomy

- Thoracic Cage
- Trachea
- Pleura
- Lung
- Diaphragm

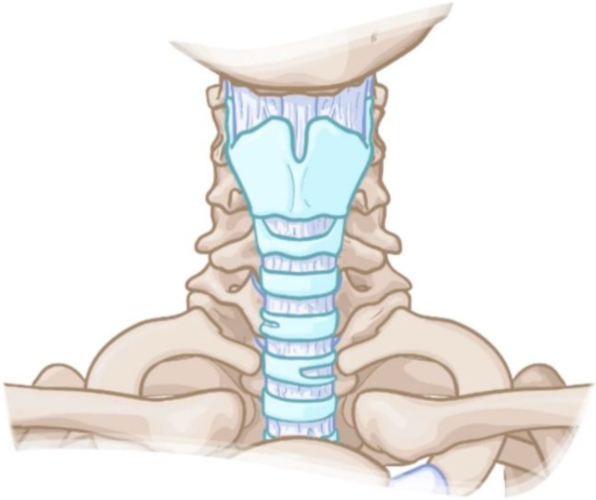


Thoracic Cage



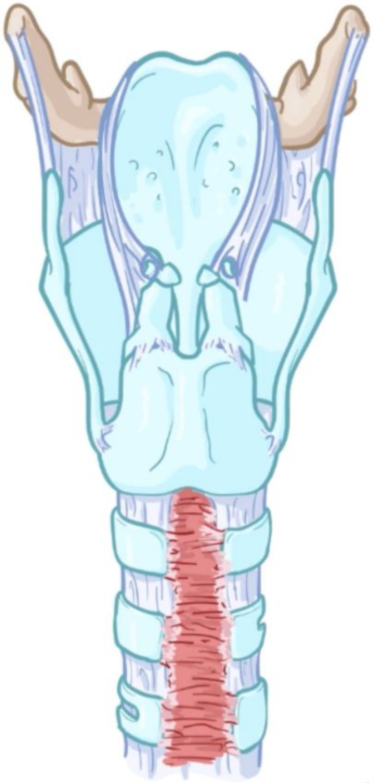
A, The Thoracic Cage—Anterior View; **B,** Sternum—Anterior View

TRACHEA



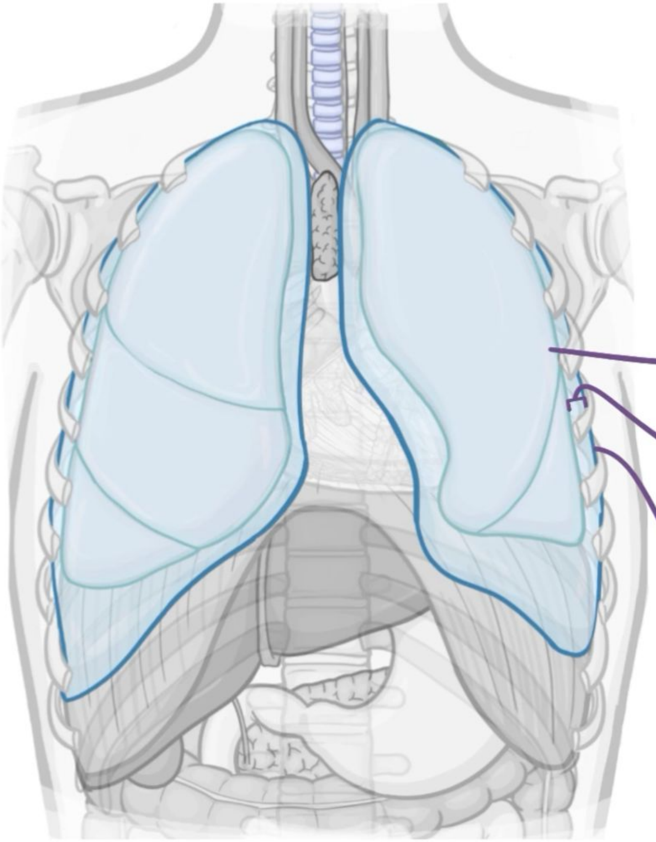
C6 → T4 - T5

* INCOMPLETE TRACHEAL CARTILAGES
* DIVIDES into RIGHT & LEFT MAIN BRONCHI



POSTERIOR

Pleura

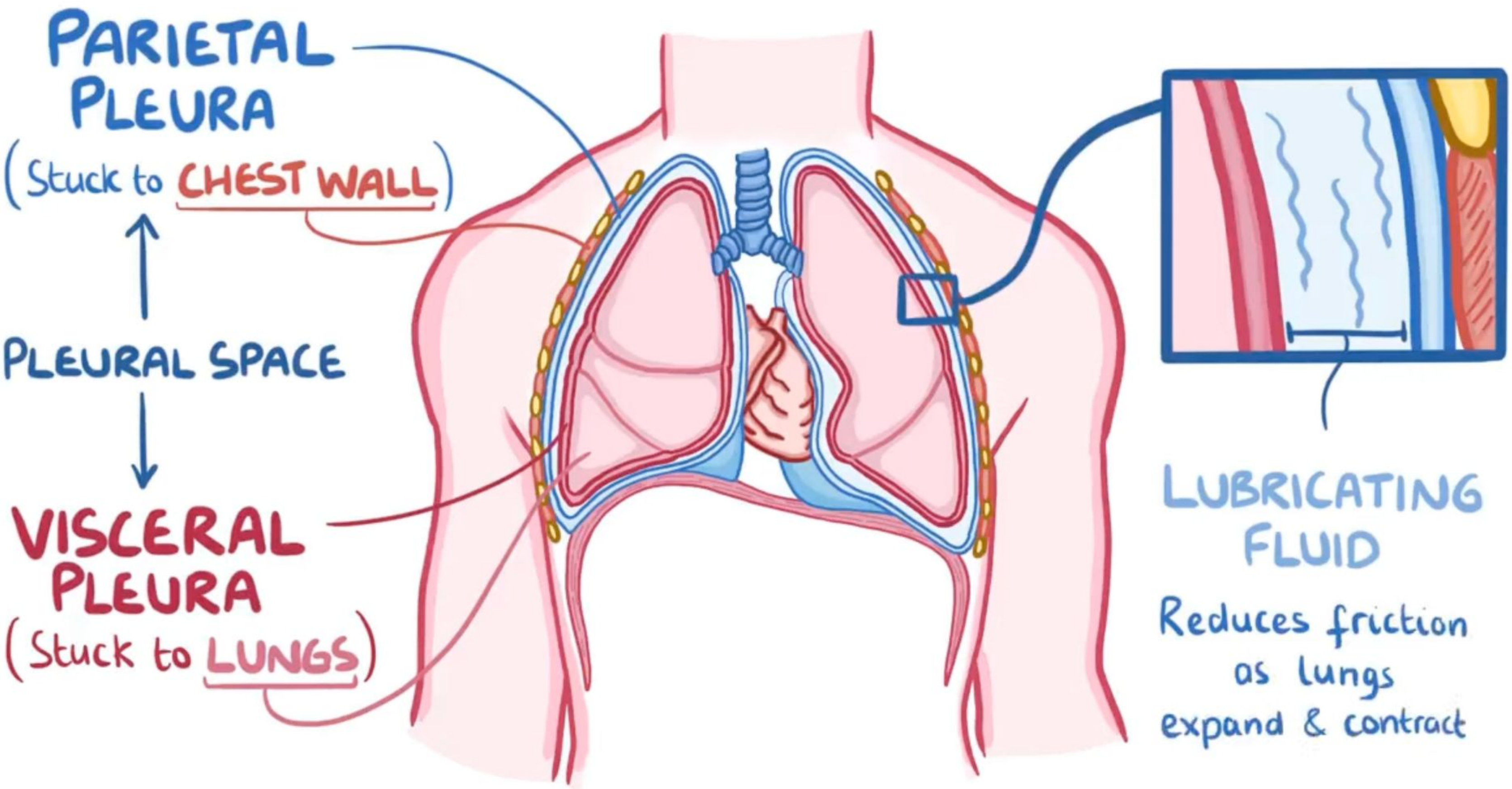


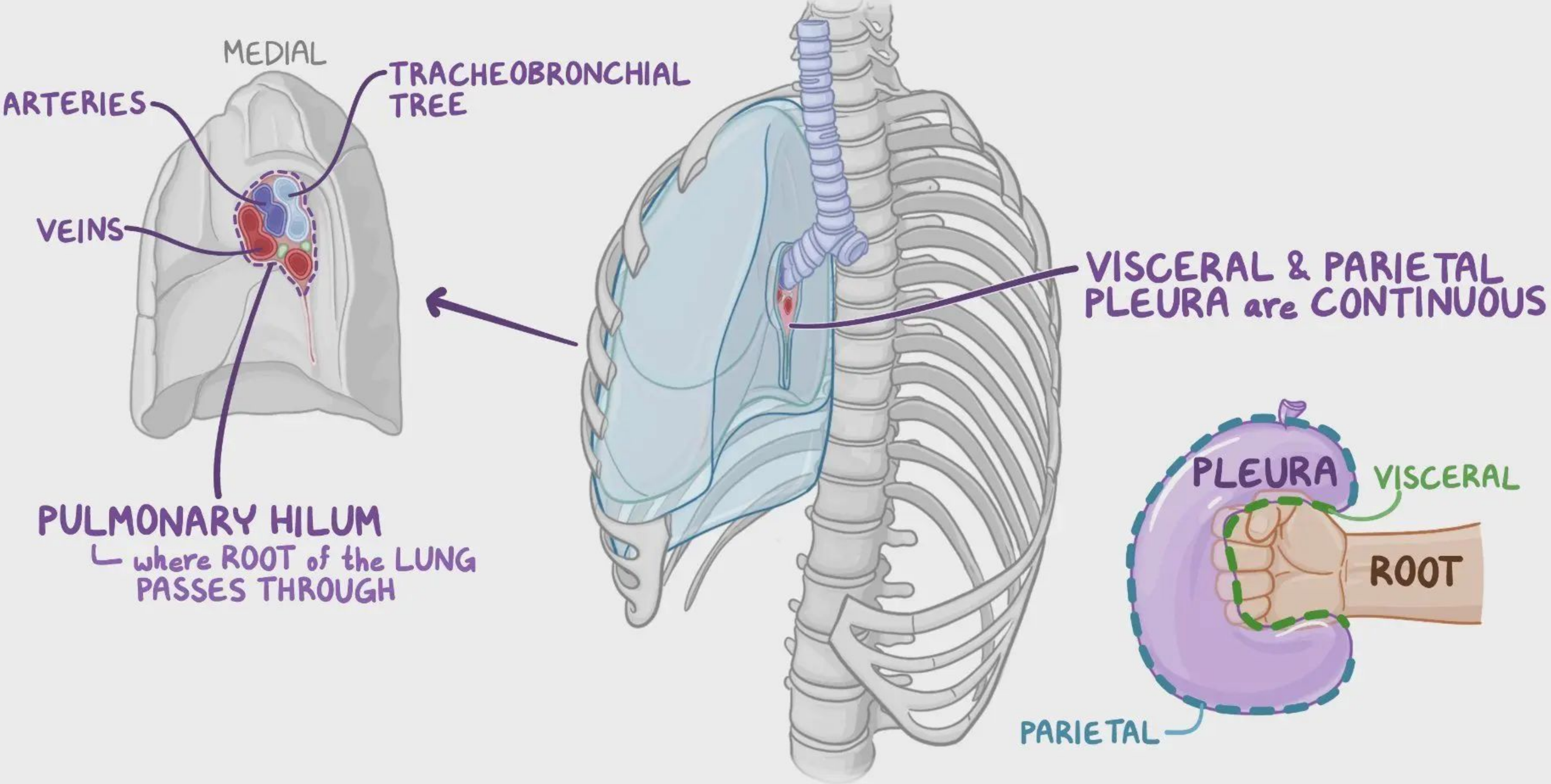
PLEURA
- THIN MEMBRANE that COVERS EACH LUNG

VISCERAL PLEURA
- INTIMATELY ADHERES to LUNG

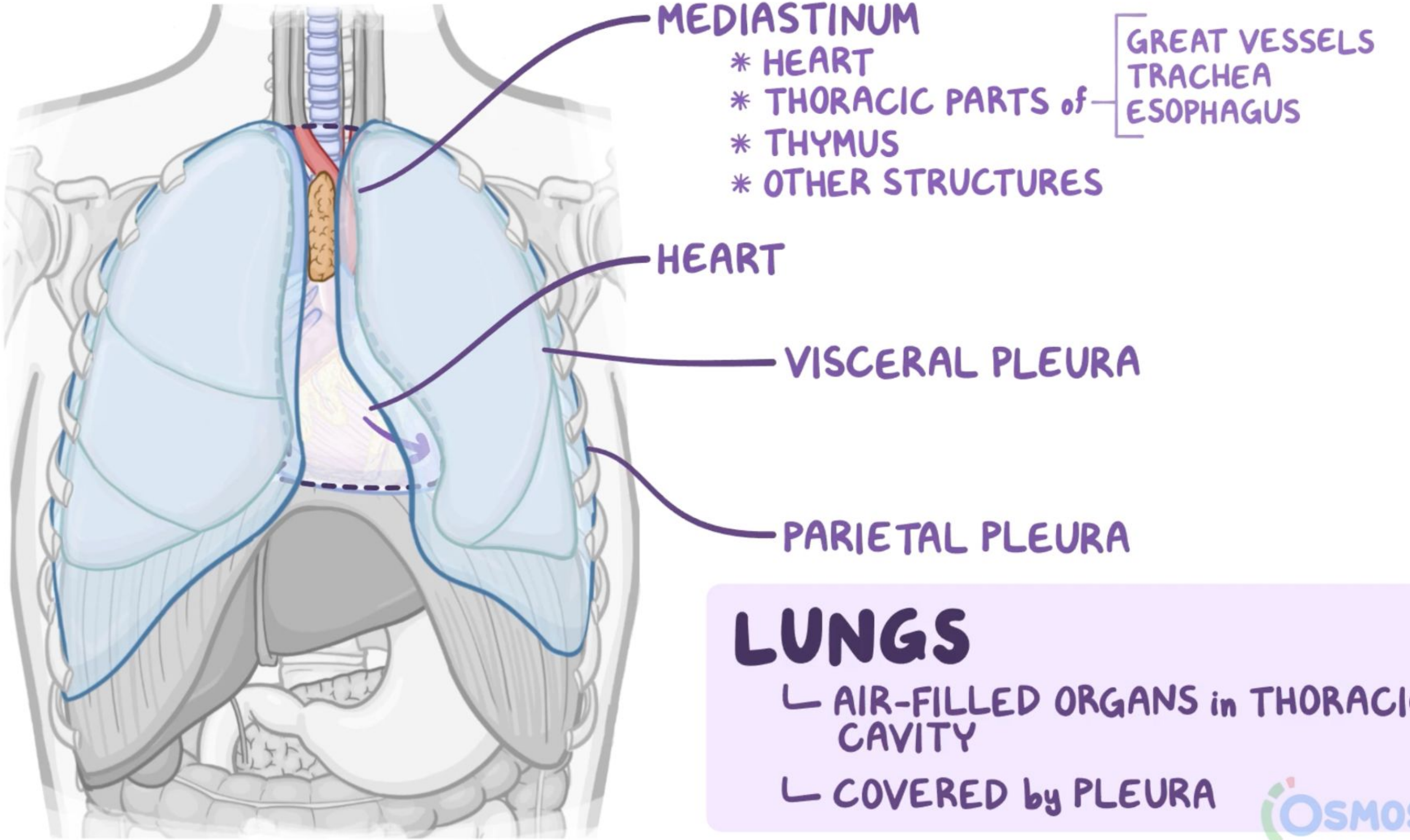
PLEURAL CAVITY
- FILLED with FLUID

PARIETAL PLEURA
- LINES PULMONARY CAVITY


PARIETAL PLEURA
(Stuck to CHEST WALL)
PLEURAL SPACE
VISCERAL PLEURA
(Stuck to LUNGS)
LUBRICATING FLUID
Reduces friction as lungs expand & contract


MEDIAL
ARTERIES
TRACHEOBRONCHIAL TREE
VEINS
PULMONARY HILUM
where ROOT of the LUNG PASSES THROUGH
VISCERAL & PARIETAL PLEURA are CONTINUOUS
PLEURA
VISCERAL
ROOT
PARIETAL

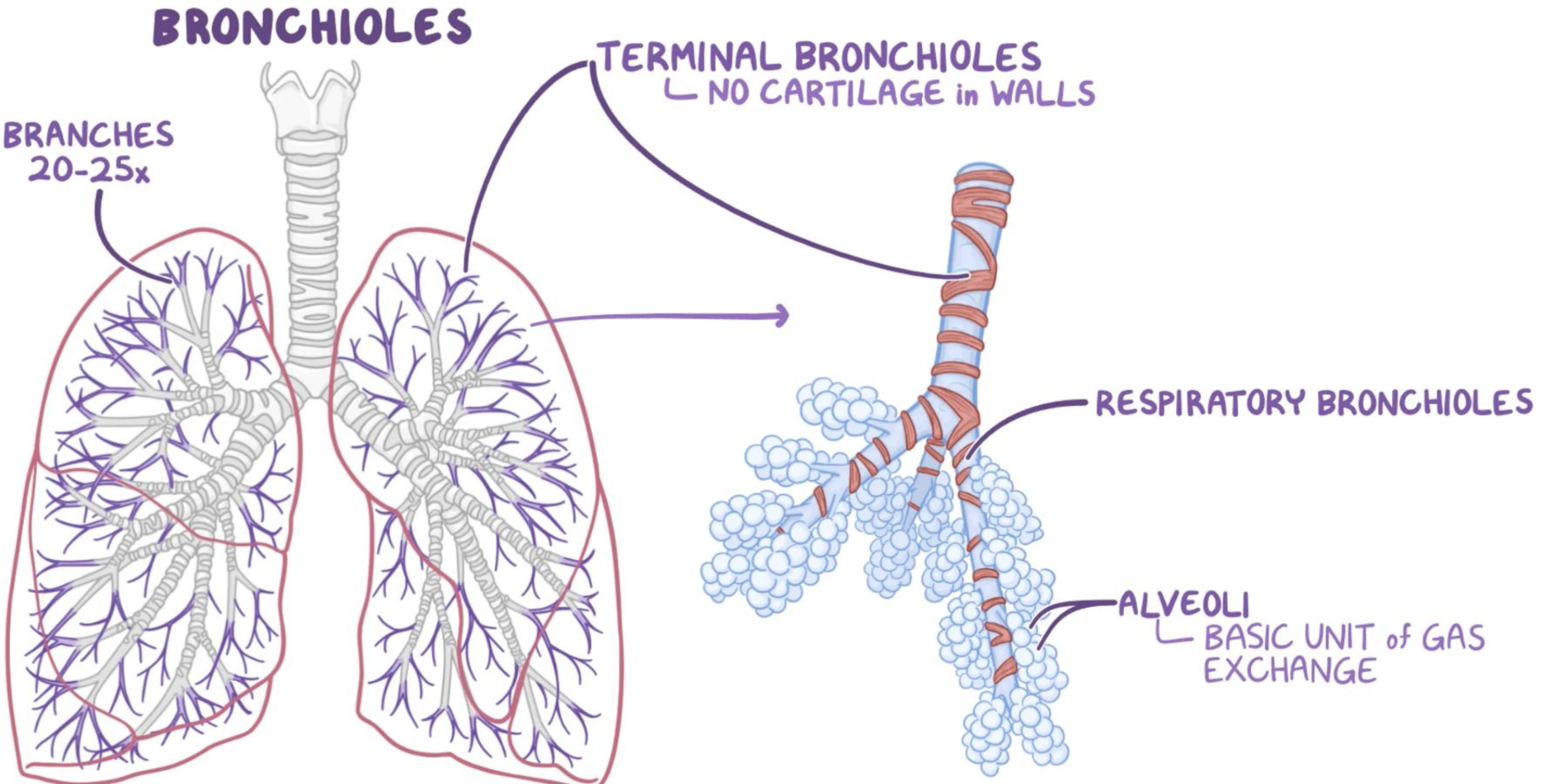
Lung

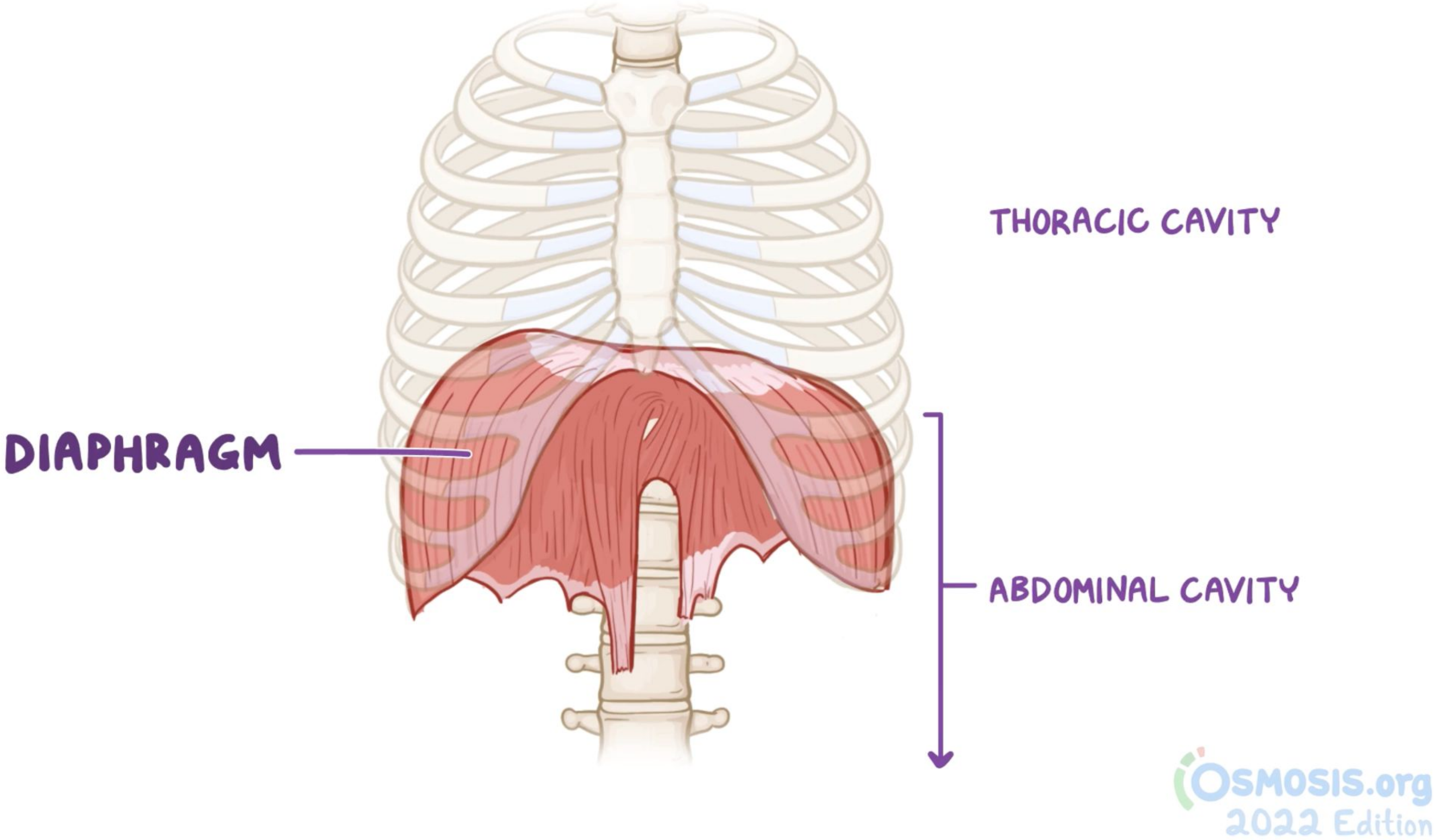


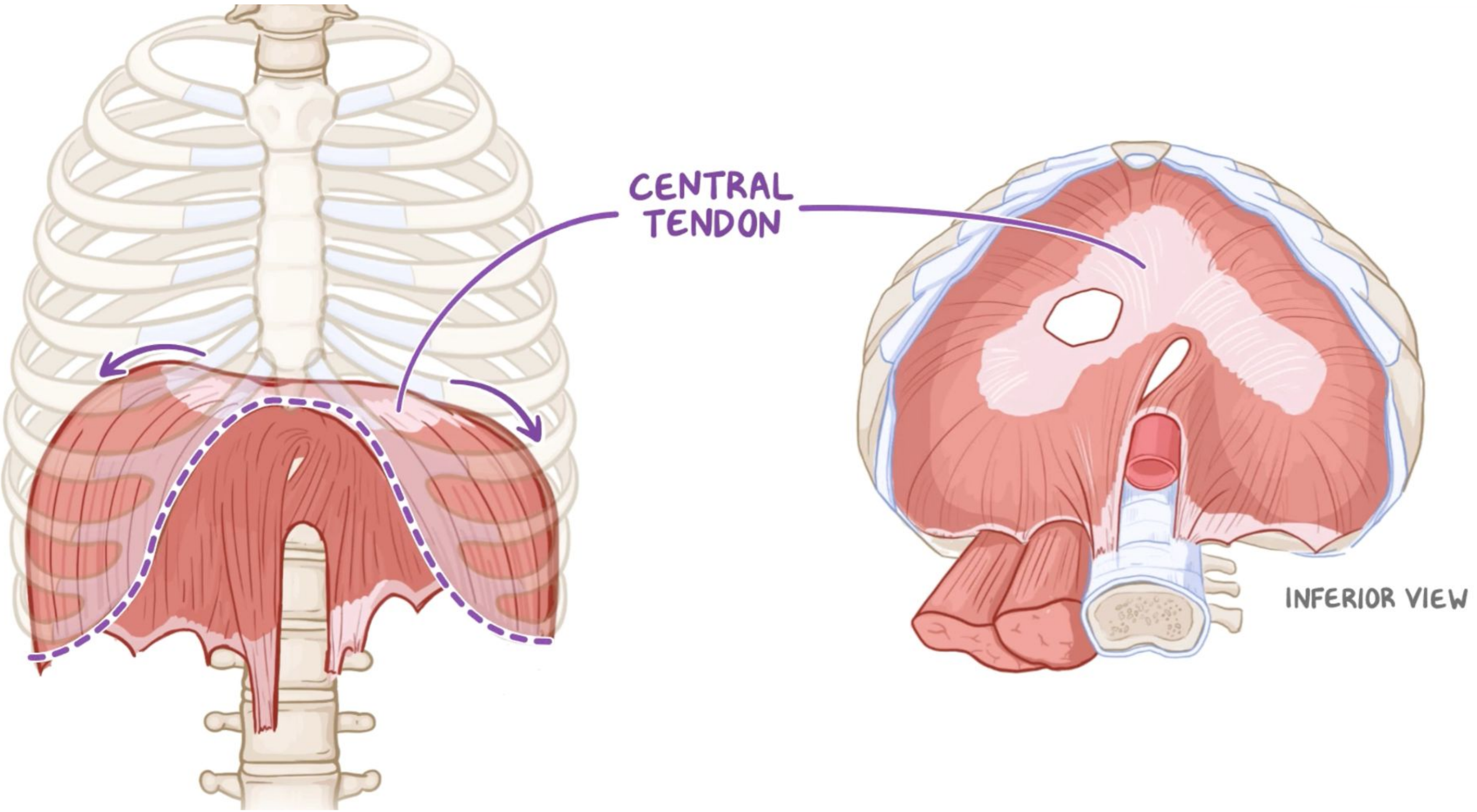
LUNGS

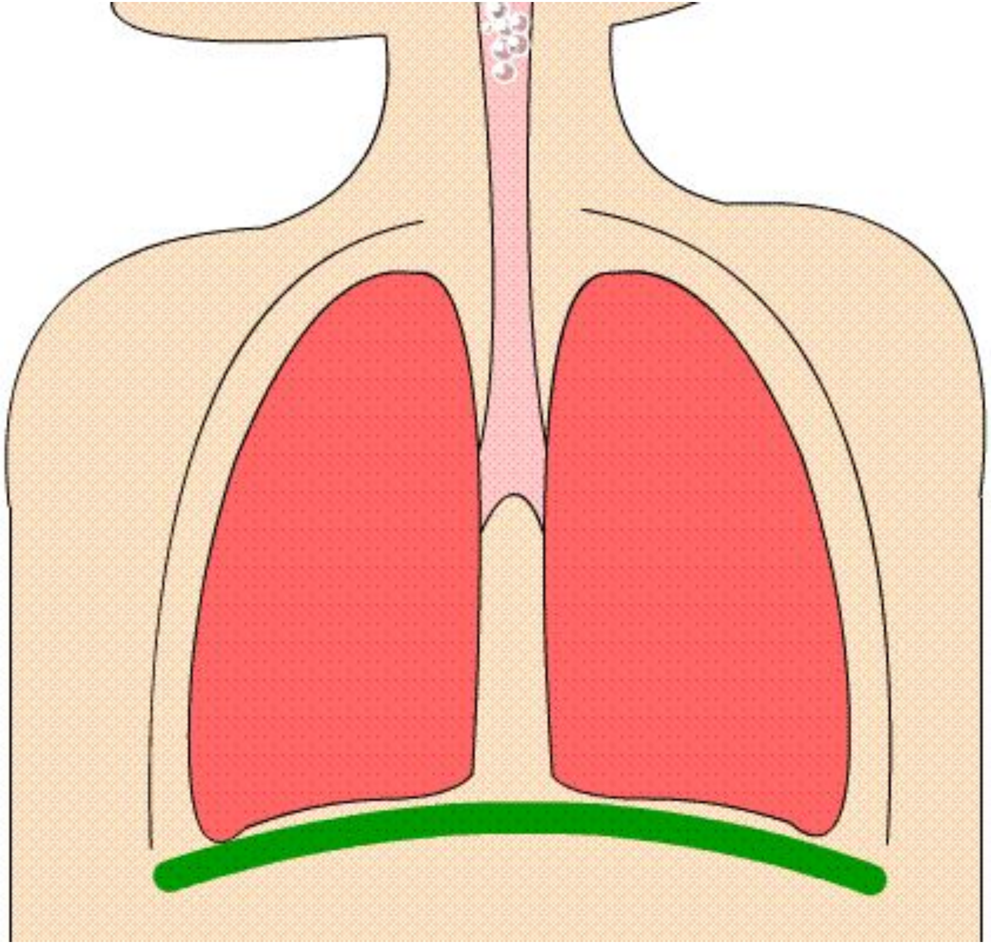
- AIR-FILLED ORGANS in THORACIC CAVITY
- COVERED by PLEURA

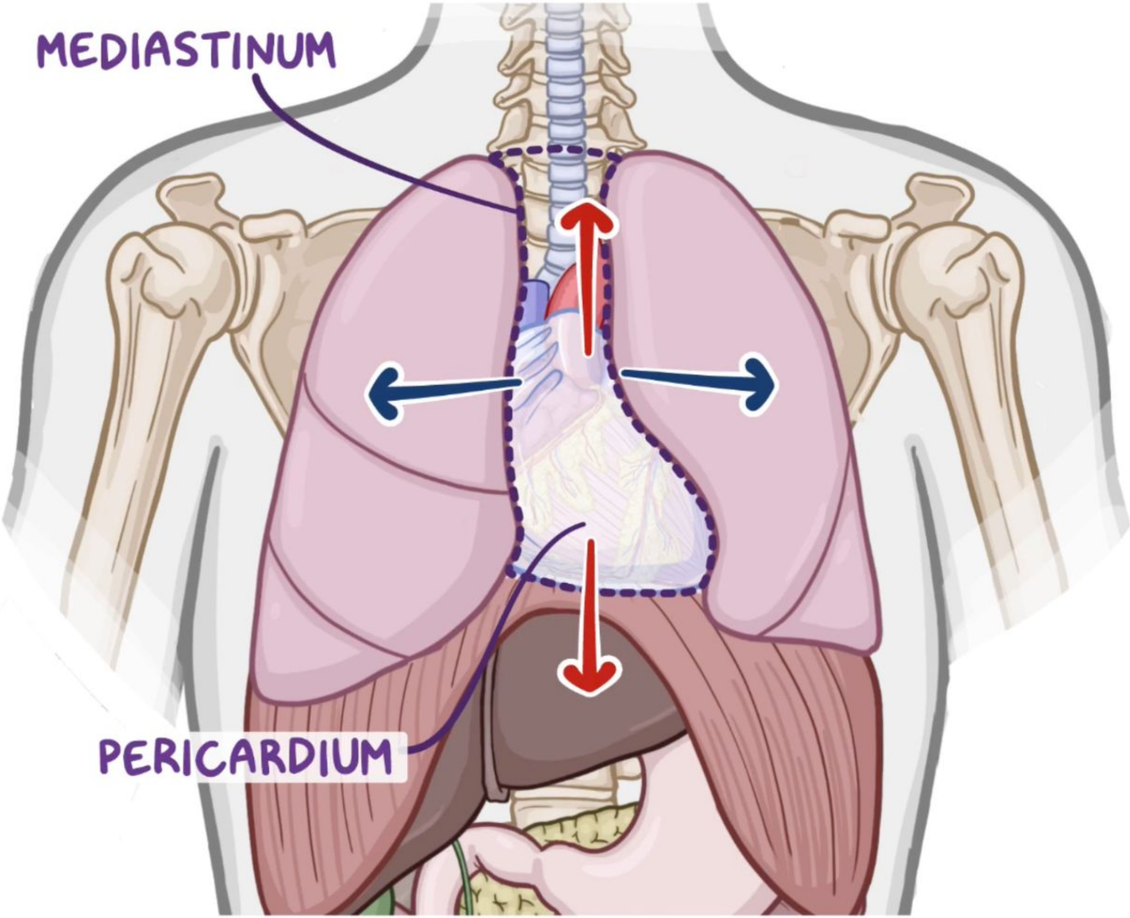
Lung


BRONCHIOLES
TERMINAL BRONCHIOLES
└ NO CARTILAGE in WALLS
BRANCHES
20-25x
RESPIRATORY BRONCHIOLES
ALVEOLI
└ BASIC UNIT of GAS
EXCHANGE

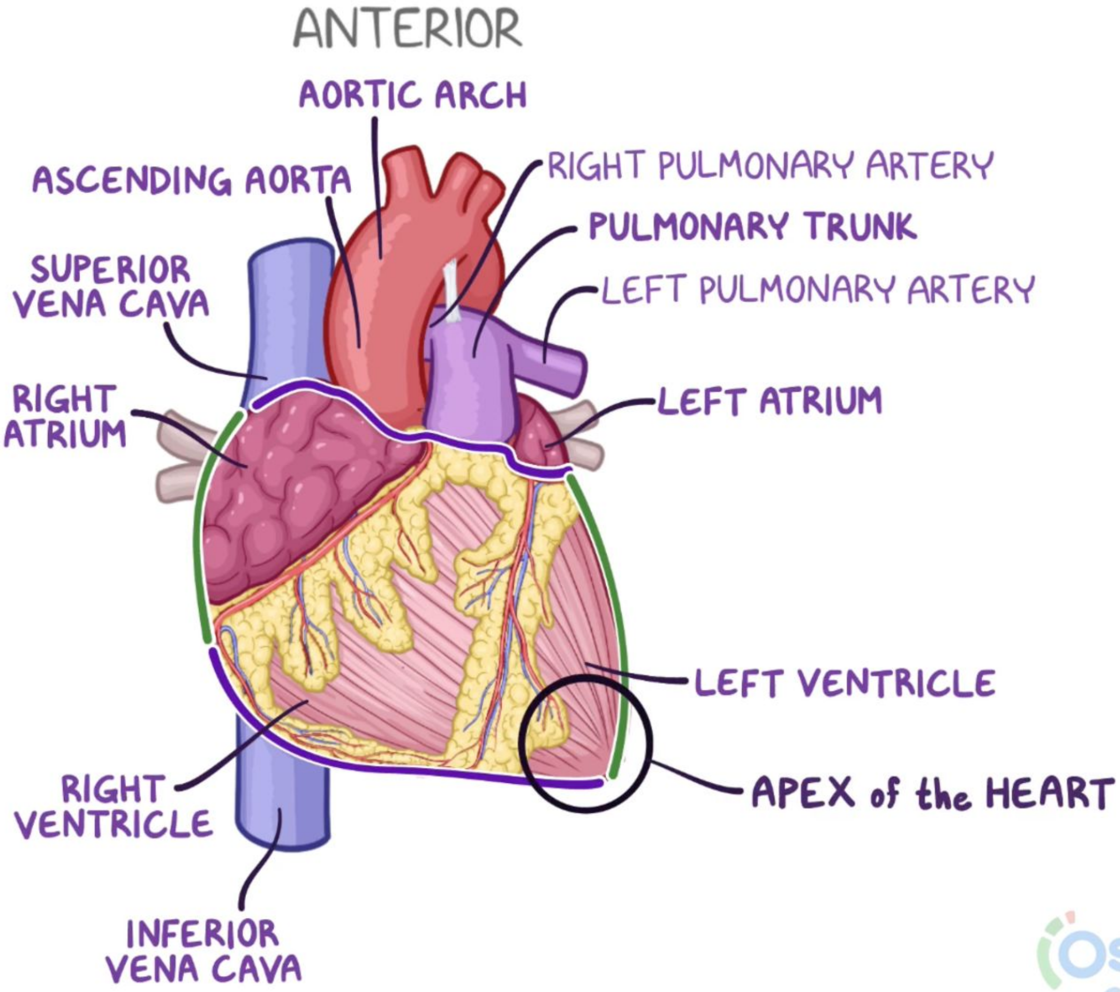

THORACIC CAVITY
DIAPHRAGM
ABDOMINAL CAVITY
OSMOSIS.org
2022 Edition

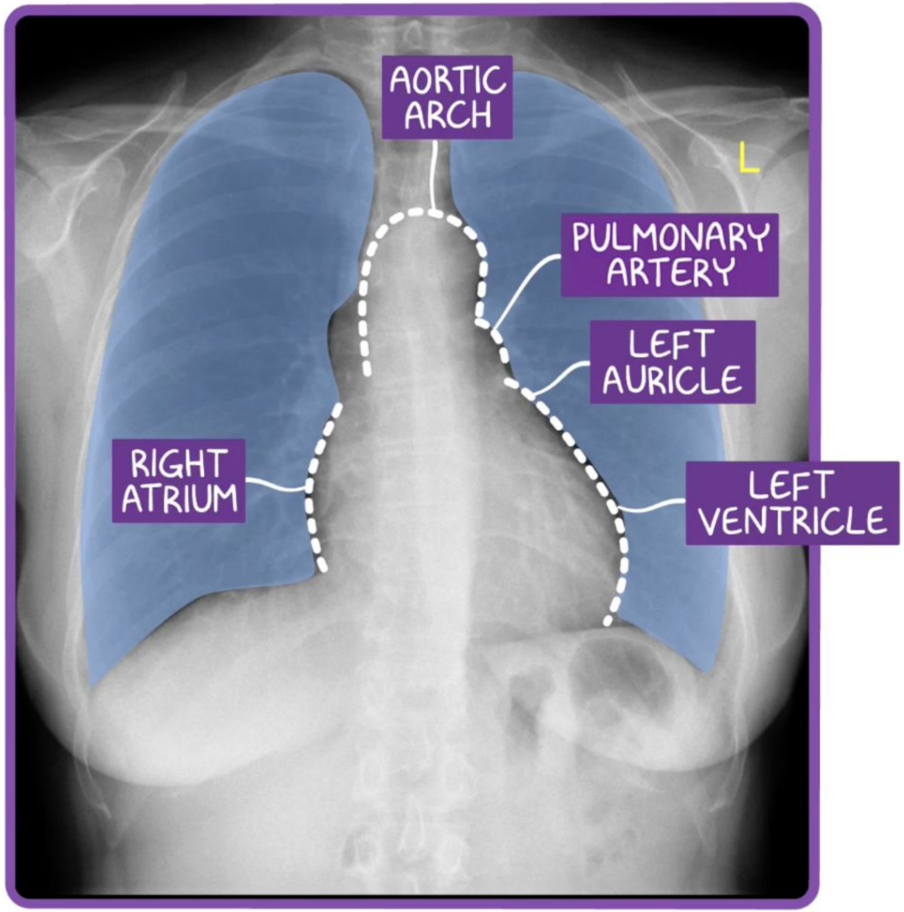

CENTRAL
TENDON
INFERIOR VIEW





MEDIASTINUM
PERICARDIUM


ANTERIOR
AORTIC ARCH
ASCENDING AORTA
RIGHT PULMONARY ARTERY
PULMONARY TRUNK
SUPERIOR VENA CAVA
LEFT PULMONARY ARTERY
RIGHT ATRIUM
LEFT ATRIUM
LEFT VENTRICLE
APEX of the HEART
RIGHT VENTRICLE
INFERIOR VENA CAVA


AORTIC ARCH
PULMONARY ARTERY
LEFT AURICLE
LEFT VENTRICLE
RIGHT ATRIUM
L

What is Chest X-Ray

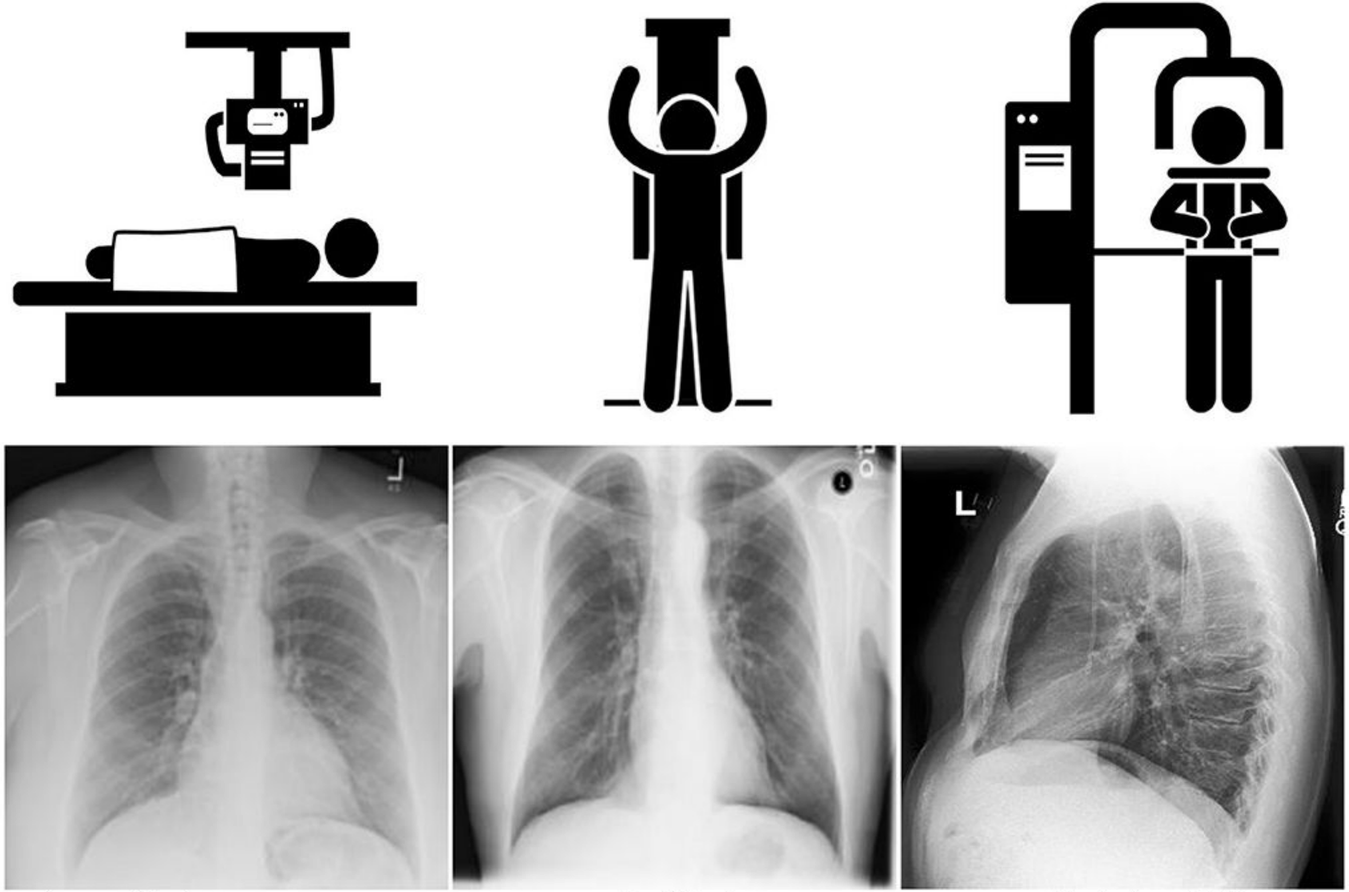
Chest X-ray is a common medical imaging test.

It uses X-rays to create pictures of the chest.

It visualizes internal organs and structures, including:

- Heart
- Lungs
- Ribs
- Diaphragm

Types Of Chest X-Ray

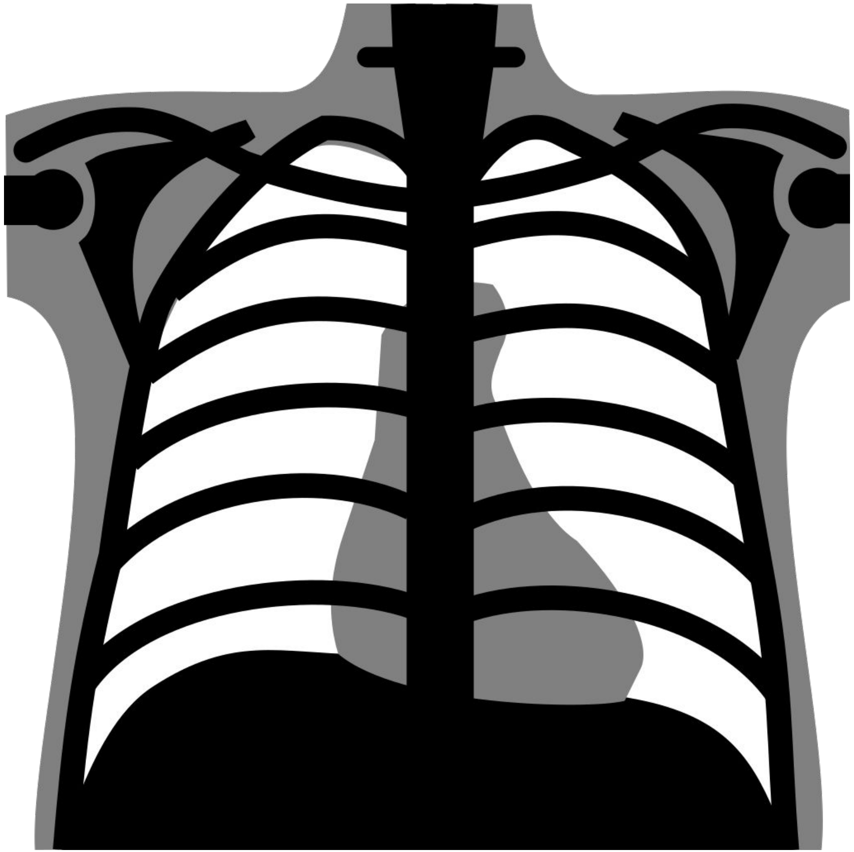


A AP view B PA view C Lateral View

TABLE 2 Illustrates the differences between two common CXR projections.

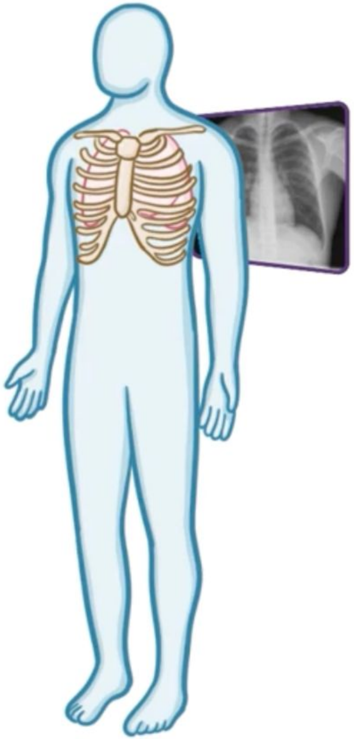
PA view	AP view
Standard frontal Chest projection	Alternative frontal projection to the PA
X-ray beam traverses the patient from posterior to anterior	X-ray beam traverses the patient from anterior to posterior
Needs full aspiration and standing position from patient	Can be performed patient sitting on the bed
Best practice to examine lungs, mediastinum and thoracic cavity	Best practice for intubated and sick patients
Heart size appear normal	Heart size appear magnified
Images are of higher quality and a better option to assess heart size	Not a good option to assess the size of heart

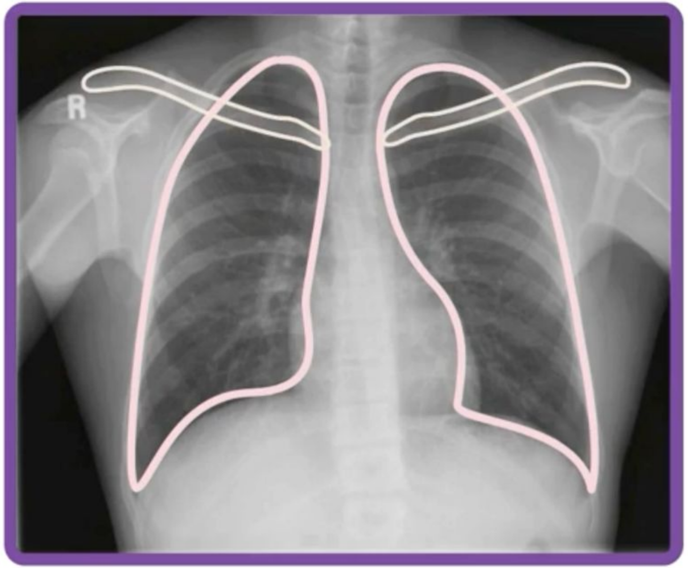
Chest X-ray Interpretation



X-RAYS

* LESS LIKELY to PENETRATE DENSER MATERIALS

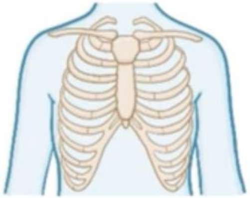




A
SSESSMENT of DATA - IMAGE QUALITY
IR WHERE it SHOULD NOT BE

E
QUIPMENT
FFUSIONS

B
ONES
ODY WALL



F
IELDS

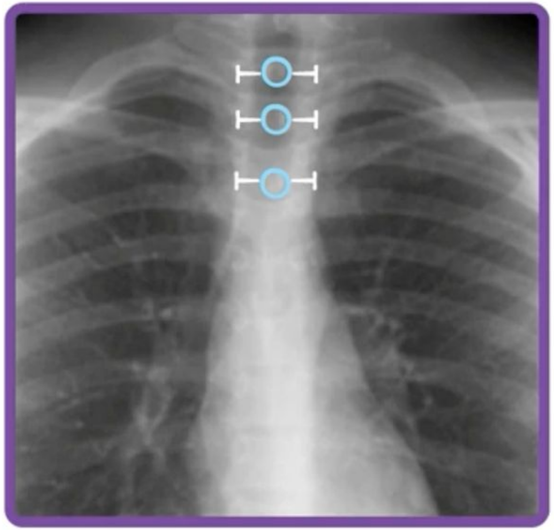
C
ARDIAC SILHOUETTE & SIZE

G
REAT VESSELS

D
IAPHRAGMS

ASSESSMENT - IMAGE QUALITY

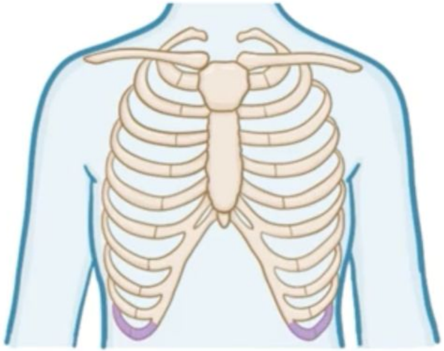
NO EXCESS ROTATION of PATIENT



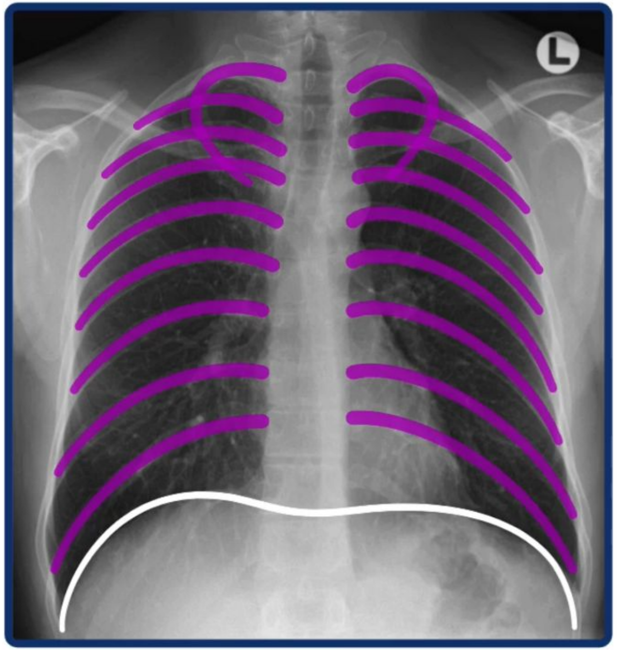
ROTATION introduces UNWANTED VARIATION

EXPIRATION

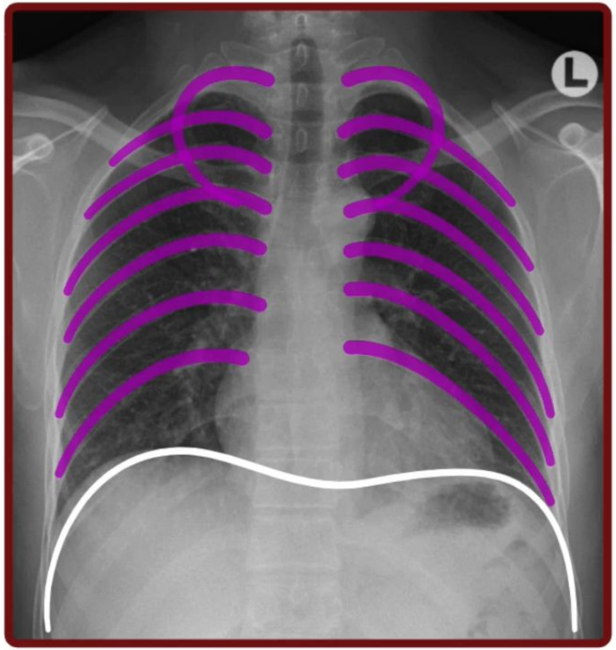
INSPIRATION



F. INSPIRATION



8-10 POSTERIOR RIBS



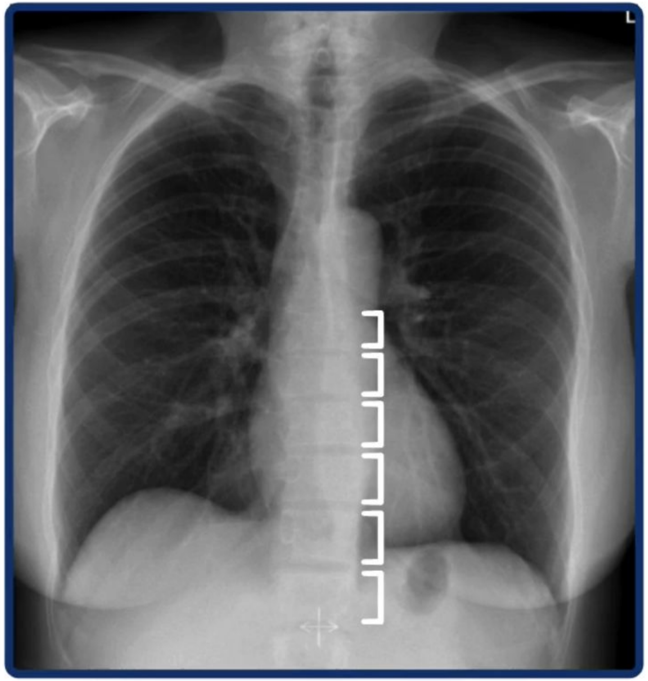
* < 8 RIBS
* CROWDED LUNG MARKINGS

ASSESSMENT - IMAGE QUALITY

E. PENETRATION

UNDERPENETRATED

OVERPENETRATED



ADEQUATELY PENETRATED

AIR WHERE it SHOULDN'T BE

SURGICAL EMERGENCY

PNEUMOTHORAX

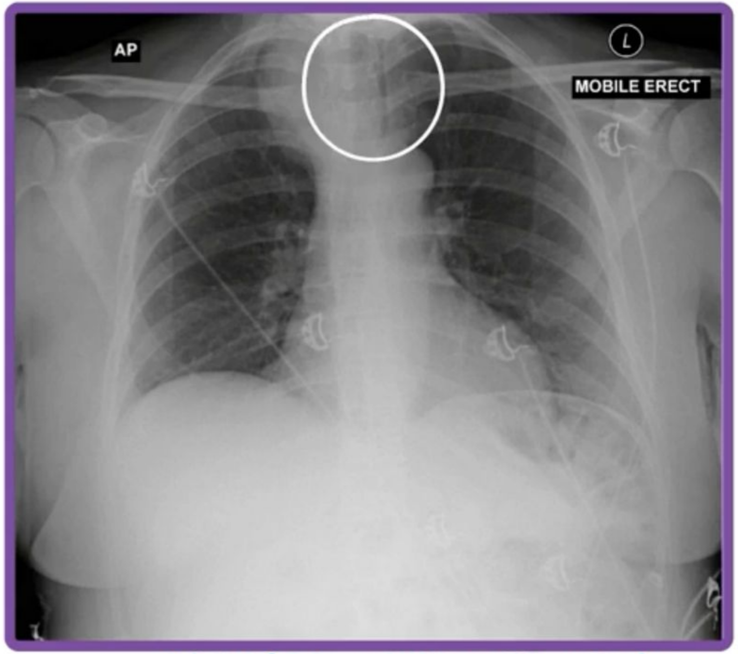
PNEUMOMEDIASTINUM

PNEUMOPERITONEUM

SUBCUTANEOUS EMPHYSEMA

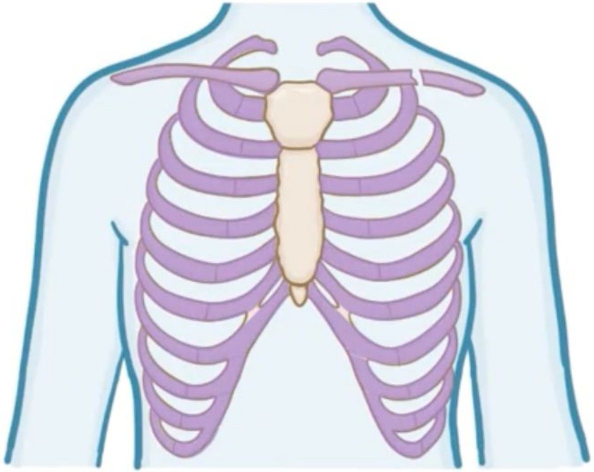
AIR WHERE it SHOULDN'T BE

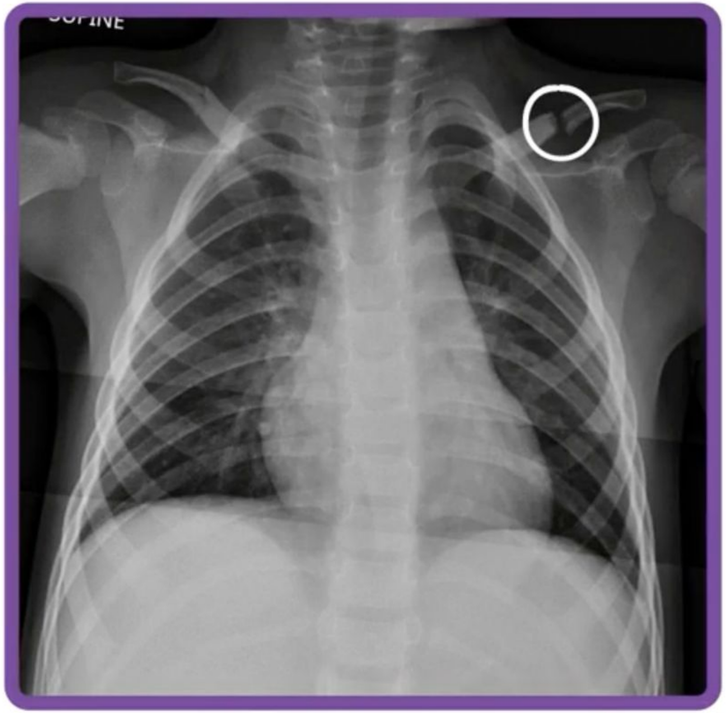
MAJOR AIRWAY BENT



may SIGNAL UNDERLYING MASS

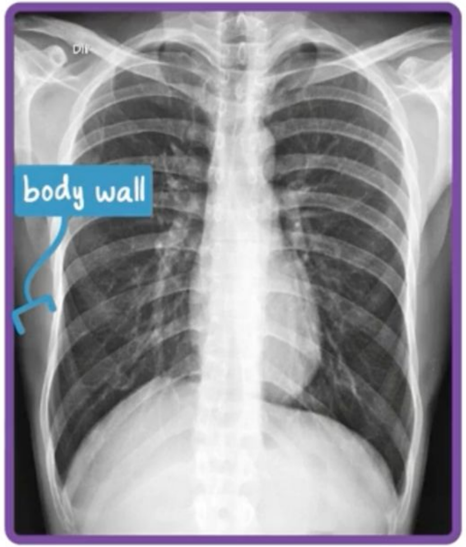
BONES

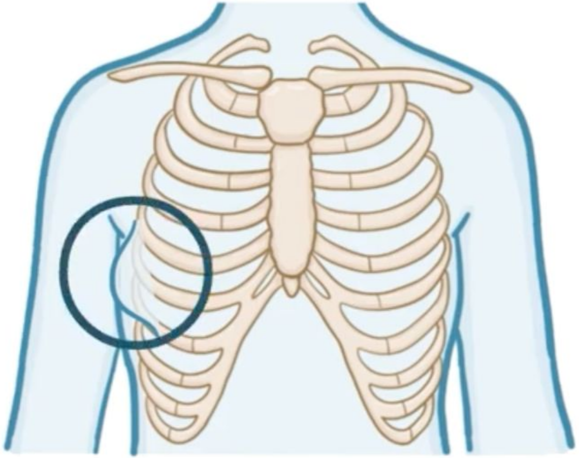




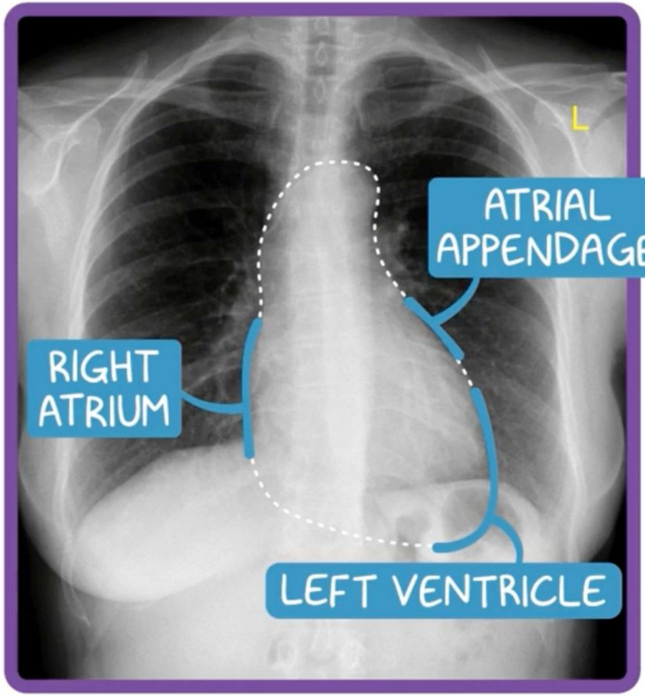
B ODY WALL

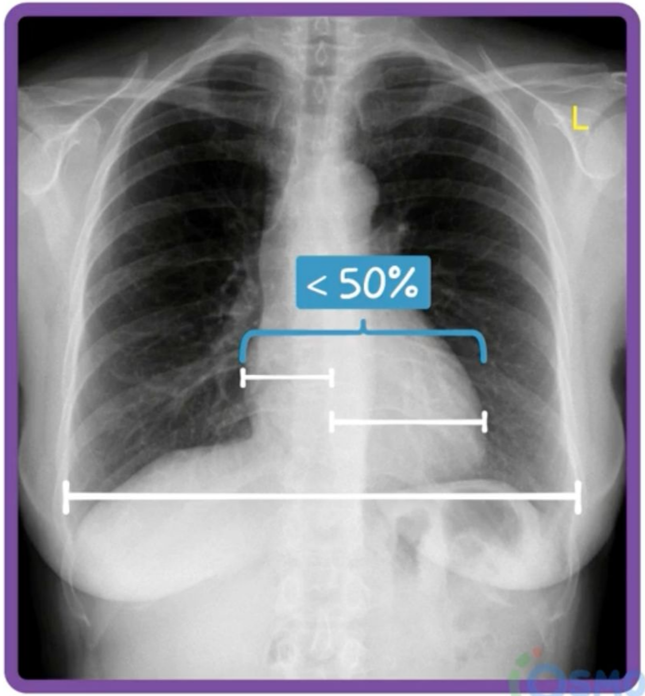
* SOFT TISSUE outside CHEST
* CHECKED for SWELLING, MASSES





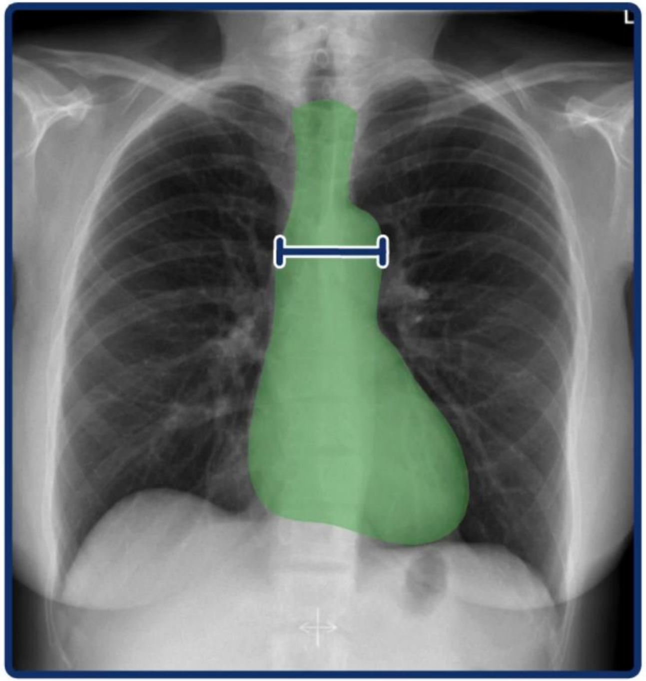
C ARDIAC SILHOUETTE & SIZE





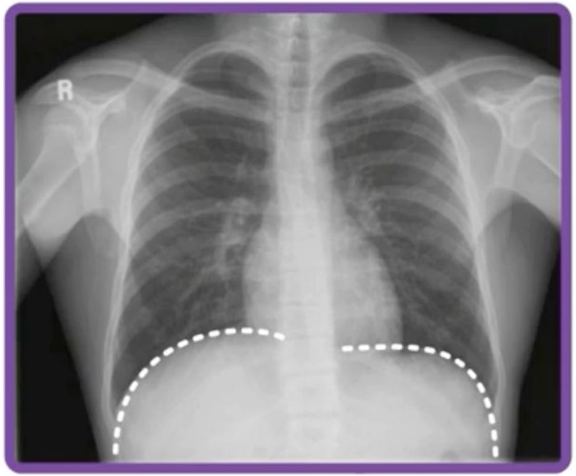
MEDIASTINUM

OVERALL WIDTH & CONTOURS of HEART & AORTA

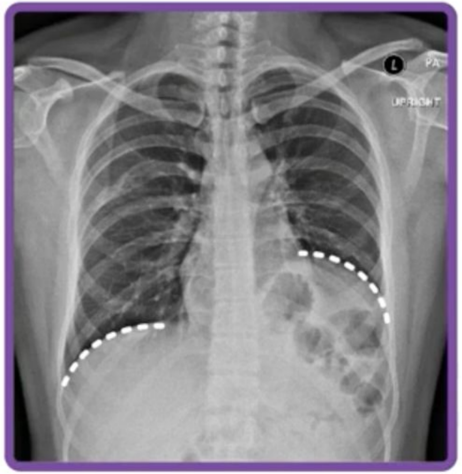


DIAPHRAGMS

SYMMETRIC



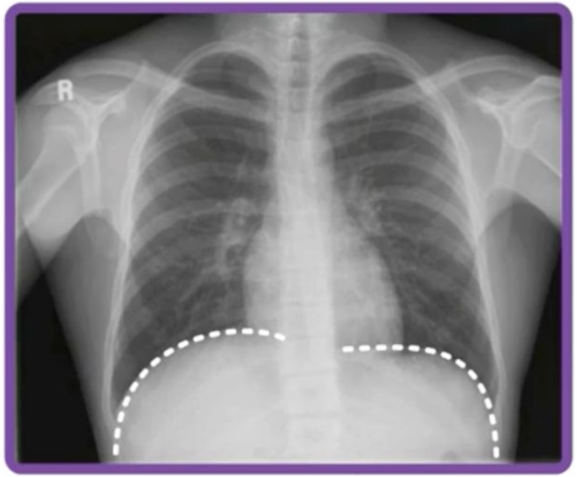
NOT SYMMETRIC



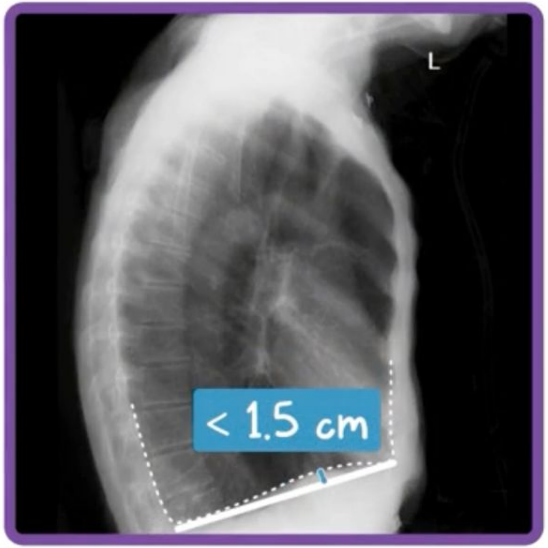
NOT TOO FLAT but FAIRLY SYMMETRIC

DIAPHRAGMS

SYMMETRIC



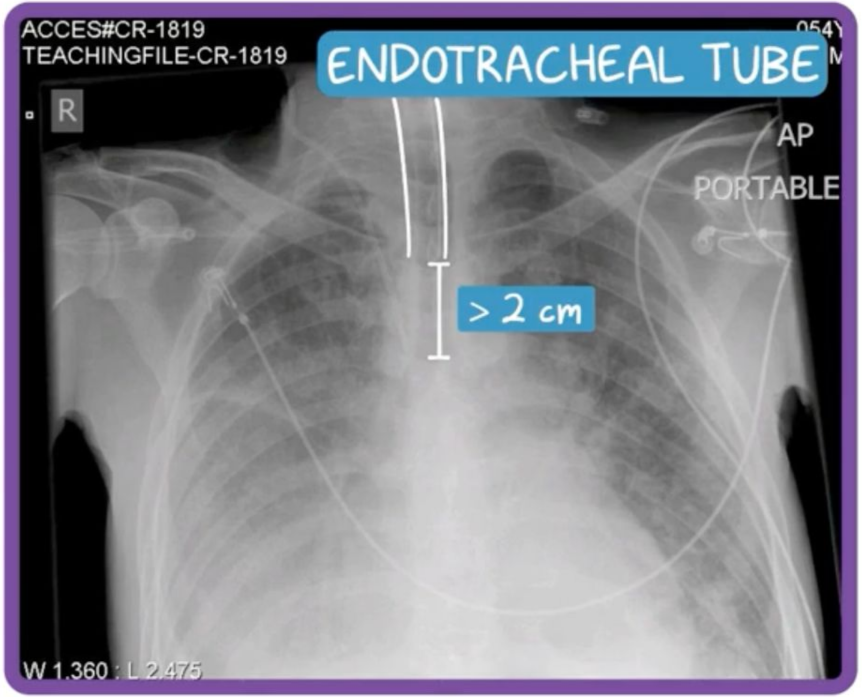
NOT SYMMETRIC

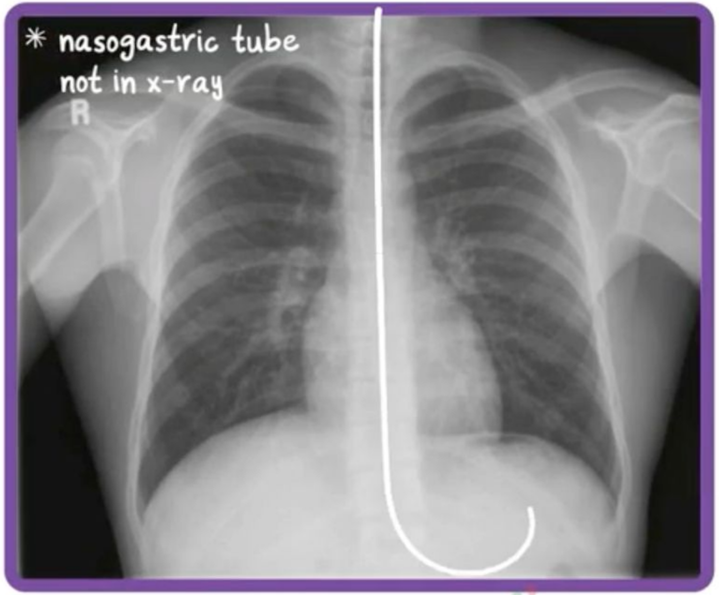


NOT TOO FLAT but FAIRLY SYMMETRIC

E QUIPMENT

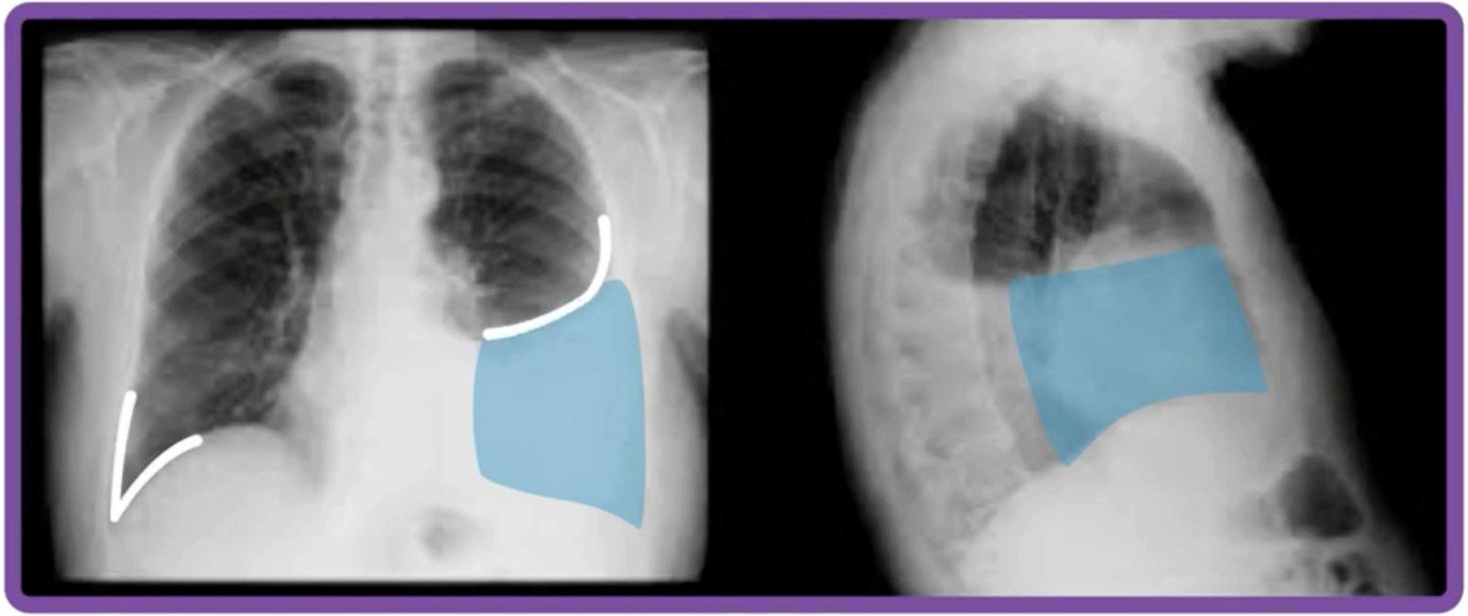
* NOTE where they are RELATIVE to OTHER STRUCTURES
* whether in FUNCTIONAL POSITION





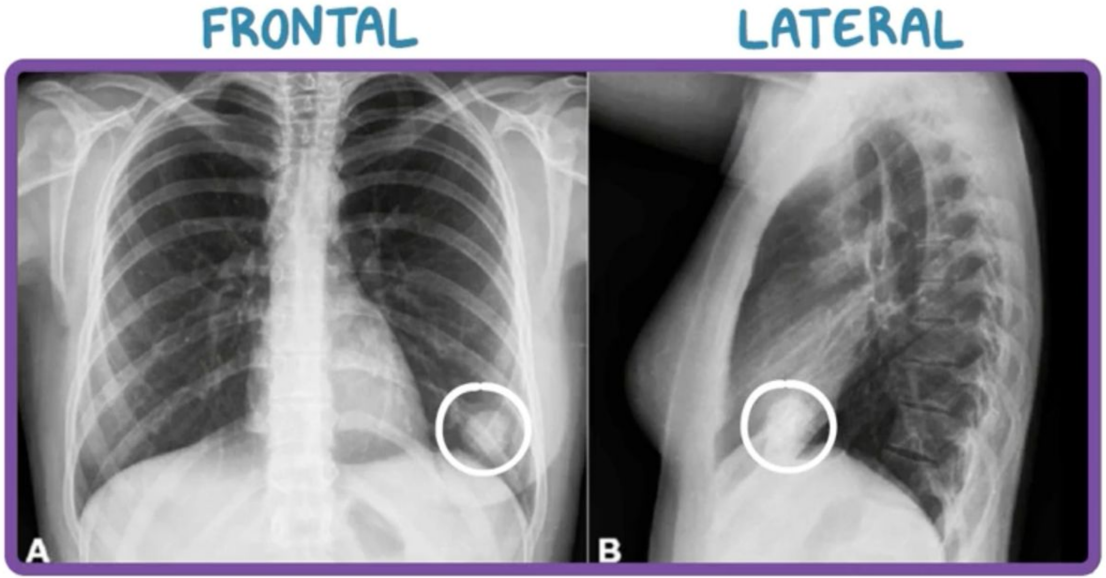
E (PLEURAL) EFFUSION

* COMMON but SUBTLE PATHOLOGY

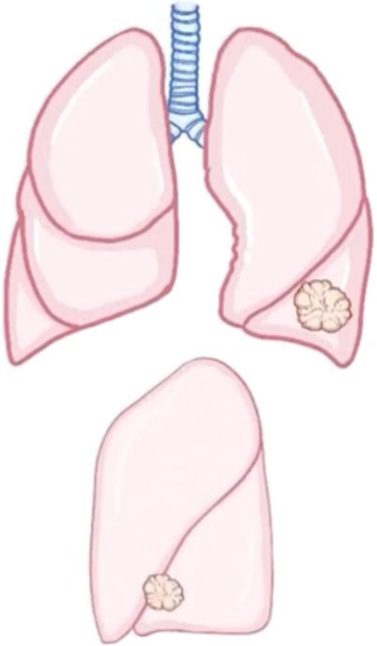


F (LUNG) FIELDS - SHOULD LOOK SYMMETRIC

* NO HAZINESS, WHITE DOTS, BLOTCHES

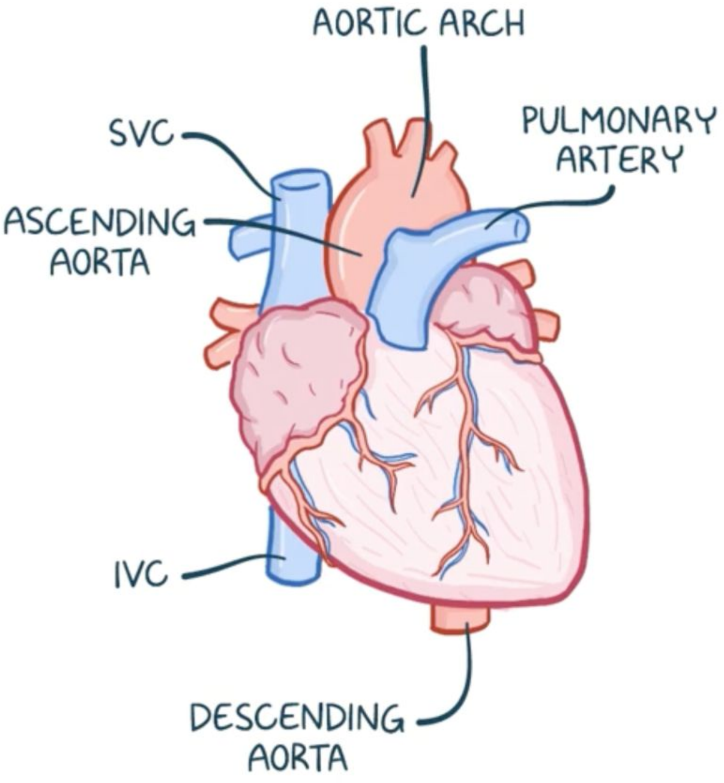


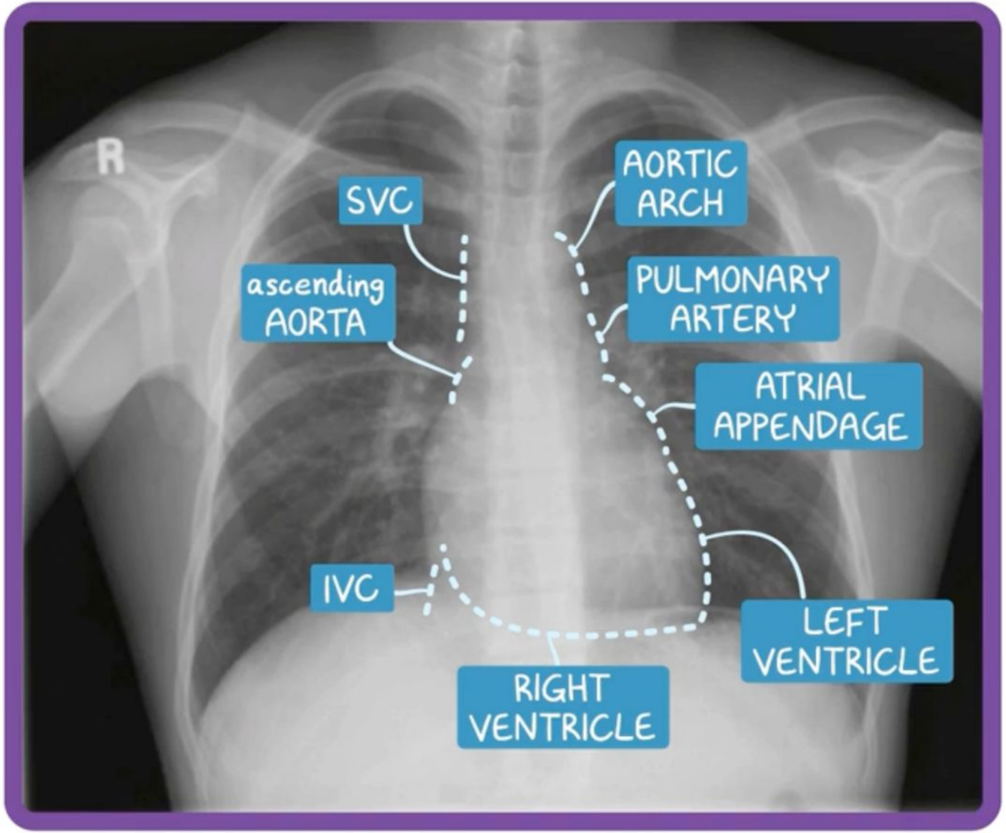
DETERMINE which LOBE & LUNG



ANTEROMEDIAL BASAL SUBSEGMENT of LEFT LOWER LOBE

GREAT VESSELS





DEVIATION ⟶ CONGENITAL ABNORMALITY or DISEASE